JEE Mathematics: Quick Review with Grading and Practice Map - Trigonometric Ratio (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (23 questions; a question that appears in two of these is counted once): High = 2 or more questions, Medium = 1 to 1 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction to Trig Ratios

Question 01

What are the six trigonometric ratios and their values in terms of P, B and H?

Model Answer

Right triangle, angle θ: $\sin\theta = \dfrac{P}{H}$, $\cos\theta = \dfrac{B}{H}$, $\tan\theta = \dfrac{P}{B}$; $\operatorname{cosec}\theta = \dfrac{H}{P}$, $\sec\theta = \dfrac{H}{B}$, $\cot\theta = \dfrac{B}{P}$.

Reciprocal pairs: $\sin\cdot\operatorname{cosec} = \cos\cdot\sec = \tan\cdot\cot = 1$; $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

Grade - Low

Practice map, main concept: Sheet: J-I 7 | DPP: 0(17)

Also used in: none

Question 02

What are the three fundamental trigonometric identities?

Model Answer

$$\sin^2\theta + \cos^2\theta = 1, \quad 1 + \tan^2\theta = \sec^2\theta, \quad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Used to eliminate θ: from $a\cos\theta + b\sin\theta = 3$, $a\sin\theta - b\cos\theta = 4$, squaring and adding gives $a^2 + b^2 = 25$.

Grade - High

Practice map, main concept: Sheet: J-I 3-4, 10, 28; J-II 39; C-I 10-11; C-II 2-3, 11 | DPP: 0(1-6, 8-14, 16, 18, 20, 22-25); 4(22) | Question Bank: 6

Also used in: Sheet: J-I 1, 18; J-M 11; C-I 14-15; C-II 8, 30 | DPP: 0(7, 15, 19); 3(26); 4(8-9, 11) | Question Bank: 2, 10

Question 03

What is the relation between $\sec\theta + \tan\theta$ and $\sec\theta - \tan\theta$?

Model Answer

$$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1, \qquad (\operatorname{cosec}\theta + \cot\theta)(\operatorname{cosec}\theta - \cot\theta) = 1$$

Ex: $\sec\theta + \tan\theta = 2 \Rightarrow \sec\theta - \tan\theta = \frac{1}{2}$, so $\sec\theta = \frac{5}{4}$, $\tan\theta = \frac{3}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 32; J-A 1 | DPP: 0(7, 15)

Also used in: DPP: 0(21)

Question 04
What is the result of $\sin^4\theta + \cos^4\theta$?

Model Answer

$$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta = 1 - \tfrac{1}{2}\sin^2 2\theta$$

Range: $\left[\tfrac{1}{2}, 1\right]$.

Grade - Low

Practice map, main concept: DPP: 0(19); 3(28); 4(18-19)

Also used in: Sheet: J-I 2; J-II 6; C-II 34 | DPP: 0(18); 3(27)

Question 05
What is the result of $\sin^6\theta + \cos^6\theta$?

Model Answer

$$\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta = 1 - \tfrac{3}{4}\sin^2 2\theta$$

Range $\left[\tfrac{1}{4}, 1\right]$. Ex: $f_k = \tfrac{1}{k}(\sin^k x + \cos^k x)$ gives $f_4 - f_6 = \tfrac{1}{4} - \tfrac{1}{6} = \tfrac{1}{12}$.

Grade - Low

Practice map, main concept: Sheet: J-I 2; J-II 9; C-I 4; C-II 30 | DPP: 3(27)

Also used in: Sheet: J-II 37; C-II 34

Question 06
What follows from $\dfrac{\sin^4\theta}{a} + \dfrac{\cos^4\theta}{b} = \dfrac{1}{a+b}$?

Model Answer

$$\sin^2\theta = \frac{a}{a+b}, \quad \cos^2\theta = \frac{b}{a+b}, \quad \frac{\sin^8\theta}{a^3} + \frac{\cos^8\theta}{b^3} = \frac{1}{(a+b)^3}$$

Reason: with $s = \sin^2\theta$ it becomes $\big((a+b)s - a\big)^2 = 0$. Ex: $10\sin^4\theta + 15\cos^4\theta = 6 \Rightarrow \sin^2\theta = \tfrac{3}{5}$, so $8\sec^6\theta + 27\operatorname{cosec}^6\theta = 250$.

Grade - Medium

Practice map, main concept: Question Bank: 2

Also used in: none

Question 07
Prove $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1 + \sin A}{\cos A}$.

Model Answer

Replace the 1 in the numerator by $\sec^2 A - \tan^2 A$: numerator $= (\tan A + \sec A)(1 - \sec A + \tan A)$, which cancels the denominator.

So LHS $= \tan A + \sec A = \dfrac{1 + \sin A}{\cos A}$.

Grade - Low

Practice map, main concept: DPP: 0(21)

Also used in: none

Question 08

What should be kept in mind while proving trigonometric identities?

Model Answer

Start from the more complicated side; convert everything to sin and cos; use $1 = \sin^2 + \cos^2 = \sec^2 - \tan^2$ to factorise; take LCM, then cancel.

Ex: $\dfrac{\tan A}{1-\cot A} + \dfrac{\cot A}{1-\tan A} = 1 + \sec A \operatorname{cosec} A$ (use $a^3 - b^3$).

Grade - Low

Practice map, main concept: Sheet: J-I 1

Also used in: DPP: 0(3-4, 6, 8-10, 14, 17, 20, 25)

Measurement of Angle

Question 09

How to convert an angle from degrees to radians and vice versa?

Model Answer

$$\pi \text{ rad} = 180^\circ, \quad 1^\circ = \frac{\pi}{180} \text{ rad}, \quad 1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57^\circ 17' 45''$$

Ex: $75^\circ = \dfrac{5\pi}{12}$; $\dfrac{7\pi}{9} = 140^\circ$. Also $1^\circ = 60'$, $1' = 60''$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 10

If no symbol is written on an angle, in which system is it taken?

Model Answer

In radians: $\sin 2$ means $\sin(2 \text{ rad}) \approx \sin 114.6^\circ > 0$. Ex: $\sin 1 > \cos 1$ since $1 \text{ rad} > \frac{\pi}{4}$.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 1(4-8)

Question 11

How to express an angle given in degrees in terms of π?

Model Answer

Multiply by $\dfrac{\pi}{180}$ and simplify: $15^\circ = \dfrac{\pi}{12}$, $18^\circ = \dfrac{\pi}{10}$, $22.5^\circ = \dfrac{\pi}{8}$, $36^\circ = \dfrac{\pi}{5}$, $120^\circ = \dfrac{2\pi}{3}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 12

Which angles are called important (quadrantal) angles?

Model Answer

Integral multiples of $\dfrac{\pi}{2}$: 0, $\dfrac{\pi}{2}$, π, $\dfrac{3\pi}{2}$, $2\pi, \ldots$ (terminal arm on an axis).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 13
How to mark quadrantal angles on the branches (axes)?

Model Answer
Anticlockwise from $+x$: 0 on $+x$, $\frac{\pi}{2}$ on $+y$, π on $-x$, $\frac{3\pi}{2}$ on $-y$, 2π back on $+x$; clockwise angles are negative ($-\frac{\pi}{2}$ on $-y$).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 14
How to mark angles other than quadrantal angles in the quadrants?

Model Answer
Remove full turns ($2n\pi$), then see between which quadrantal angles it lies. Ex: $\frac{17\pi}{6} = 2\pi + \frac{5\pi}{6}$ lies in Q-II; $-\frac{\pi}{3}$ lies in Q-IV.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 29

Question 15
What is the sign of all trigonometric ratios in each quadrant?

Model Answer
Q-I: all $+$; Q-II: $\sin, \operatorname{cosec} +$; Q-III: $\tan, \cot +$; Q-IV: $\cos, \sec +$ (rest negative) - "All Students Take Coffee".

Grade - Medium
Practice map, main concept: Sheet: C-I 14; C-II 23 | DPP: 1(9-14)
Also used in: Sheet: J-A 1 | DPP: 4(23)

Reduction Formulae

Question 16
How to apply reduction formulae?

Model Answer
Write the angle as $n\frac{\pi}{2} \pm \theta$ (θ acute). n odd: sin $\leftrightarrow$ cos, tan $\leftrightarrow$ cot, sec $\leftrightarrow$ cosec; n even: ratio unchanged.

Sign = sign of the original ratio in the quadrant of $n\frac{\pi}{2} \pm \theta$. Ex: $\cos\frac{5\pi}{3} = \cos\left(2\pi - \frac{\pi}{3}\right) = \frac{1}{2}$.

Grade - Medium
Practice map, main concept: DPP: 2(7)
Also used in: Sheet: J-A 4; C-II 3 | DPP: 1(15-27)

Question 17
Write the reduction formulae for $\left(\frac{\pi}{2} - \theta\right)$.

Model Answer

$$\sin\left(\tfrac{\pi}{2} - \theta\right) = \cos\theta,\ \cos\left(\tfrac{\pi}{2} - \theta\right) = \sin\theta,\ \tan\left(\tfrac{\pi}{2} - \theta\right) = \cot\theta$$

For $\frac{\pi}{2} + \theta$: $\sin \to \cos\theta,\ \cos \to -\sin\theta,\ \tan \to -\cot\theta$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 6; C-II 26 | DPP: 2(3-4, 7, 18, 33); 3(13); 4(18-19)

Question 18
Write the reduction formulae for $(\pi - \theta)$.

Model Answer

$$\sin(\pi - \theta) = \sin\theta,\ \cos(\pi - \theta) = -\cos\theta,\ \tan(\pi - \theta) = -\tan\theta$$

For $\pi + \theta$: $\sin \to -\sin\theta,\ \cos \to -\cos\theta,\ \tan \to \tan\theta$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 4(18-19); 5(14)

Question 19
Write the reduction formulae for $(-\theta)$.

Model Answer

$$\sin(-\theta) = -\sin\theta,\ \cos(-\theta) = \cos\theta,\ \tan(-\theta) = -\tan\theta$$

$\cos, \sec$ are even; $\sin, \tan, \cot, \operatorname{cosec}$ are odd. $2\pi - \theta$ behaves like $-\theta$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 20
What are the values of sine and cosine at each quadrantal angle?

Model Answer
$\sin(n\pi) = 0,\ \cos(n\pi) = (-1)^n$; $\sin\left((2n+1)\frac{\pi}{2}\right) = (-1)^n,\ \cos\left((2n+1)\frac{\pi}{2}\right) = 0$.
At $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$: $\sin = 0, 1, 0, -1$; $\cos = 1, 0, -1, 0$.

Grade - Low
Practice map, main concept: Sheet: J-I 23
Also used in: none

Question 21
What to do when two complementary angles are observed?

Model Answer
$A + B = \dfrac{\pi}{2}$: $\sin A = \cos B,\ \tan A = \cot B$, so $\tan A \tan B = 1$; pair such terms.
Ex: $\tan 1^\circ \tan 2^\circ \cdots \tan 89^\circ = 1$; $\sin^2 10^\circ + \sin^2 80^\circ = 1$.

Grade - **High**
Practice map, main concept: Sheet: J-M 5; C-II 4
Also used in: Sheet: J-I 5, 22; J-II 9, 15, 34; J-M 4; C-I 6, 9; C-II 5, 15 | Question Bank: 4

Question 22
What to do when two supplementary angles are observed?

Model Answer
$A + B = \pi$: $\sin A = \sin B,\ \cos A = -\cos B,\ \tan A = -\tan B$; terms cancel or double.
Ex: $\cos 1^\circ + \cos 2^\circ + \cdots + \cos 179^\circ = 0$ (pair k° with $(180 - k)^\circ$).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 23
How to check the sign of $\sin A + \cos A$ for a given A?

Model Answer

$$\sin A + \cos A = \sqrt{2} \sin\left(A + \tfrac{\pi}{4}\right)$$

Positive for $-\dfrac{\pi}{4} < A < \dfrac{3\pi}{4}$, negative for $\dfrac{3\pi}{4} < A < \dfrac{7\pi}{4}$ (zero on the line $y = -x$).

Grade - **Low**
Practice map, main concept: DPP: 1(28-31)
Also used in: Sheet: C-I 1 | DPP: 4(23-24)

Question 24
How to check the sign of $\sin A - \cos A$ for a given A?

Model Answer

$$\sin A - \cos A = \sqrt{2} \sin\left(A - \tfrac{\pi}{4}\right)$$

Positive for $\dfrac{\pi}{4} < A < \dfrac{5\pi}{4}$, negative for $-\dfrac{3\pi}{4} < A < \dfrac{\pi}{4}$ (zero on $y = x$).

Grade - **Low**
Practice map, main concept: DPP: 1(32-34)
Also used in: DPP: 4(23-24)

Question 25

How to compute all remaining trig ratios at A when one ratio at A is given?

Model Answer

Draw a right triangle with the given ratio, find the third side by Pythagoras, then attach signs by the quadrant of A.

Ex: $\tan A = -\dfrac{5}{12}$, A in Q-II: $\sin A = \dfrac{5}{13}$, $\cos A = -\dfrac{12}{13}$.

Grade - High

Practice map, main concept: Sheet: J-II 33; J-M 8 | DPP: 4(2)

Also used in: Sheet: J-I 17; J-II 39; J-A 6; C-II 18, 28 | DPP: 2(1-2); 3(1); 4(1, 3-5) | Question Bank: 9

Question 26

How to convert a trigonometric ratio of a higher angle into a ratio of an acute angle?

Model Answer

Remove multiples of 360° (period), then use $180^\circ \pm \theta$ or $90^\circ \pm \theta$ with the quadrant sign.

Ex: $\sin 1230^\circ = \sin 150^\circ = \dfrac{1}{2}$; $\tan(-585^\circ) = -\tan 225^\circ = -1$.

Grade - High

Practice map, main concept: Sheet: J-I 22; J-II 13; J-M 1; C-I 9 | DPP: 1(1-3, 5-8, 15-27)

Also used in: Sheet: J-I 23; J-II 15; J-M 2; C-II 10 | DPP: 1(28-34)

Compound Angles

Question 27

Write all 20 formulae of the compound angle heading.

Model Answer

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}, \quad \cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B), \quad 2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B), \quad 2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$\sin C \pm \sin D = 2 \sin \frac{C \pm D}{2} \cos \frac{C \mp D}{2}$, $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$, $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

Ex: $\tan 50^\circ - \tan 40^\circ = 2 \tan 10^\circ$ (from $\tan(50^\circ - 40^\circ)$ with $\tan 50^\circ \tan 40^\circ = 1$).

Grade - High

Practice map, main concept: Sheet: J-I 6, 12-13, 15, 17-18; J-II 3, 7-8, 10, 42; J-M 7; J-A 5; C-I 13, 16; C-II 6, 9, 17, 28 | DPP: 2(1-4, 6, 8-18, 20, 27-28, 30-35); 3(14, 18, 24-25, 34); 4(1, 8-11, 14, 21) | Question Bank: 4, 8-9

Also used in: Sheet: J-I 5, 16, 19; J-II 1, 11-12, 16-17, 21, 33, 36; J-M 5-6; J-A 6; C-I 6, 19; C-II 7, 32 | DPP: 1(1); 2(5, 19, 21); 3(15, 23, 32-33, 40); 4(3-4, 16-17, 20); 5(1, 3-7, 10-12)

Question 28

When to use $\sin(A+B)\sin(A-B)$ and $\cos(A+B)\cos(A-B)$?

Model Answer

$$\sin(A+B)\sin(A-B)=\sin^2 A-\sin^2 B,\quad \cos(A+B)\cos(A-B)=\cos^2 A-\sin^2 B$$

Use when a difference of squares of sines/cosines appears. Ex: $\sin^2\dfrac{5\pi}{12}-\sin^2\dfrac{\pi}{12}=\sin\dfrac{\pi}{2}\sin\dfrac{\pi}{3}=\dfrac{\sqrt{3}}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-II 1, 36; C-II 1 | DPP: 3(23); 4(16, 20)

Also used in: Sheet: C-I 13 | DPP: 4(14, 21); 5(9)

Question 29

What is the use of remembering $\tan\left(\dfrac{\pi}{4}+A\right)$ and $\tan\left(\dfrac{\pi}{4}-A\right)$?

Model Answer

$$\tan\left(\tfrac{\pi}{4}+A\right)=\frac{1+\tan A}{1-\tan A}=\frac{\cos A+\sin A}{\cos A-\sin A},\quad \tan\left(\tfrac{\pi}{4}-A\right)=\frac{1-\tan A}{1+\tan A}$$

Any ratio $\dfrac{1\pm\tan A}{1\mp\tan A}$ or $\dfrac{\cos A\pm\sin A}{\cos A\mp\sin A}$ becomes one tangent. Ex: $\dfrac{\cos 11^\circ+\sin 11^\circ}{\cos 11^\circ-\sin 11^\circ}=\tan 56^\circ$.

Grade - Low

Practice map, main concept: Sheet: J-I 16; J-II 14 | DPP: 3(16); 4(15)

Also used in: Sheet: J-I 11 | DPP: 3(11)

Question 30

What to keep in mind in proving identities with several terms in numerator and denominator?

Model Answer

Pair terms whose angles have equal sums/differences and apply $C\pm D$; a common factor appears in numerator and denominator and cancels.

Ex: $\dfrac{\sin A+\sin 3A+\sin 5A}{\cos A+\cos 3A+\cos 5A}=\dfrac{\sin 3A(2\cos 2A+1)}{\cos 3A(2\cos 2A+1)}=\tan 3A.$

Grade - Low

Practice map, main concept: Sheet: J-I 8 | DPP: 2(19, 21-26, 29); 3(10, 21-22)

Also used in: none

2A Formulae

Question 31

What is the formula for $\sin 2A$ in sin and cos, and what is its application?

Model Answer

$$\sin 2A=2\sin A\cos A$$

Use to combine a product $\sin A\cos A$ (repeatedly in products of cosines) or to write $\sin A=2\sin\frac{A}{2}\cos\frac{A}{2}$. Ex: $\sin 15^\circ\cos 15^\circ=\frac{1}{4}$.

Grade - Medium

Practice map, main concept: Sheet: C-II 5, 13 | DPP: 3(1, 17, 19, 38)

Also used in: Sheet: J-I 8, 22; J-II 7, 13, 33, 42; J-A 5; C-II 1, 6-7, 15-16, 28-29 | DPP: 2(20); 3(4-5, 7-9, 12, 15, 20, 24-26, 30-31); 4(1-2, 18-19, 22); 5(13)

Question 32
What is the formula for $\sin 2A$ in tan, and what is its application?

Model Answer

$$\sin 2A = \frac{2\tan A}{1+\tan^2 A}$$

Use when $\tan A$ is given or to write everything in $t = \tan A$. Ex: $\tan A = \frac{1}{2} \Rightarrow \sin 2A = \frac{4}{5}$.

Grade - Low
Practice map, main concept: Sheet: J-II 11 | DPP: 4(12)
Also used in: Sheet: J-II 41 | DPP: 3(1, 3)

Question 33
What is the formula for $\cos 2A$ in sin and cos, and what is its application?

Model Answer

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

Use to remove squares: $\sin^2 A = \dfrac{1-\cos 2A}{2}$, $\cos^2 A = \dfrac{1+\cos 2A}{2}$. Ex: $\sin A = \frac{3}{5} \Rightarrow \cos 2A = \frac{7}{25}$.

Grade - High
Practice map, main concept: Sheet: J-II 37 | DPP: 3(2, 15, 30, 37, 39)
Also used in: Sheet: J-II 33, 38; J-A 4-5; C-I 3, 12; C-II 13, 28-29 | DPP: 3(4-5, 8, 17, 19, 23, 27-28, 31, 38); 4(14); 5(1, 3)

Question 34
What is the formula for $\cos 2A$ in tan, and what is its application?

Model Answer

$$\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$$

Use when $\tan A$ is known. Ex: $\tan A = \frac{1}{2} \Rightarrow \cos 2A = \frac{3}{5}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 41; J-A 3 | DPP: 3(3, 13); 4(13)
Also used in: Sheet: C-II 22 | DPP: 3(2); 4(5-6)

Question 35
What to keep in mind when $1+\sin$ or $1-\sin$ is observed?

Model Answer

$$1 \pm \sin A = \left(\cos\tfrac{A}{2} \pm \sin\tfrac{A}{2}\right)^2$$

So $\sqrt{1 \pm \sin A} = \left|\cos\frac{A}{2} \pm \sin\frac{A}{2}\right|$ - decide the sign from the range of A. Also $\dfrac{1+\sin A}{\cos A} = \tan\left(\dfrac{\pi}{4} + \dfrac{A}{2}\right)$.

Grade - Low
Practice map, main concept: Sheet: J-I 11; J-II 38 | DPP: 3(11)
Also used in: DPP: 4(15)

Question 36
What to keep in mind when $1+\cos$ or $1-\cos$ is observed?

Model Answer

$$1+\cos A = 2\cos^2\tfrac{A}{2}, \qquad 1-\cos A = 2\sin^2\tfrac{A}{2}$$

$\sqrt{2+2\cos A} = 2\left|\cos\tfrac{A}{2}\right|$ (nested: $\sqrt{2+\sqrt{2+2\cos 8\theta}} = 2\cos 2\theta$ for small θ); $\dfrac{1-\cos A}{\sin A} = \tan\dfrac{A}{2}$.

Grade - Low
Practice map, main concept: Sheet: C-I 8, 17 | DPP: 3(9, 12, 20, 26, 29)
Also used in: Sheet: J-I 21; J-II 9; C-II 1, 17 | DPP: 3(6, 10, 21-22); 4(10, 16-17); 5(6-8, 10-12)

Question 37
What is the formula for $\tan 2A$ in tan, and what is its application?

Model Answer

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

Use to go from $\tan A$ to $\tan 2A$, or (with the quadratic $t^2\tan 2A + 2t - \tan 2A = 0$) back. Ex: $\tan A = \frac{1}{3} \Rightarrow \tan 2A = \frac{3}{4}$.

Grade - Low
Practice map, main concept: Sheet: C-II 22, 25
Also used in: Sheet: J-I 20; J-II 2; C-II 27 | DPP: 3(16)

Question 38
How to calculate the value at angle A when values at $2A$ are known?

Model Answer

$$\sin\tfrac{A}{2} = \pm\sqrt{\frac{1-\cos A}{2}}, \quad \cos\tfrac{A}{2} = \pm\sqrt{\frac{1+\cos A}{2}}, \quad \tan\tfrac{A}{2} = \frac{1-\cos A}{\sin A}$$

Sign from the quadrant of $\frac{A}{2}$. Ex: $\cos A = \frac{7}{25}$, $A \in (0,\pi)$: $\sin\frac{A}{2} = \frac{3}{5}$, $\cos\frac{A}{2} = \frac{4}{5}$.

Grade - Low
Practice map, main concept: DPP: 4(3-7)
Also used in: DPP: 4(2)

Question 39
What are $\tan A$ and $\tan^2 A$ in terms of $\sin 2A$ and $\cos 2A$?

Model Answer

$$\tan A = \frac{1-\cos 2A}{\sin 2A} = \frac{\sin 2A}{1+\cos 2A}, \qquad \tan^2 A = \frac{1-\cos 2A}{1+\cos 2A}$$

Ex: $\tan\dfrac{\pi}{8} = \dfrac{1-\cos\frac{\pi}{4}}{\sin\frac{\pi}{4}} = \sqrt{2}-1$.

Grade - Low
Practice map, main concept: Sheet: J-II 4 | DPP: 3(4-6)
Also used in: Sheet: C-I 17; C-II 21 | DPP: 4(8)

Question 40
What are the values of all trig ratios at $15°, 30°, 45°, 60°, 75°$ and $90°$?

Model Answer

$$\sin 15° = \cos 75° = \frac{\sqrt{3}-1}{2\sqrt{2}},\ \cos 15° = \sin 75° = \frac{\sqrt{3}+1}{2\sqrt{2}},\ \tan 15° = 2-\sqrt{3},\ \tan 75° = 2+\sqrt{3}$$

$\sin 30° = \frac{1}{2},\ \sin 45° = \frac{1}{\sqrt{2}},\ \sin 60° = \frac{\sqrt{3}}{2},\ \sin 90° = 1$ (cos in reverse order); $\tan 30° = \frac{1}{\sqrt{3}},\ \tan 45° = 1,\ \tan 60° = \sqrt{3}$, $\tan 90°$ undefined.

Grade - Low
Practice map, main concept: Sheet: J-I 14, 26, 29 | DPP: 1(4)
Also used in: Sheet: J-II 4-5; C-II 4, 9, 23 | DPP: 1(5-14); 2(6, 32); 3(18); 4(7)

Question 41
What are the values of tan and cot at $22.5°$ and $67.5°$?

Model Answer

$$\tan 22.5° = \cot 67.5° = \sqrt{2}-1, \qquad \tan 67.5° = \cot 22.5° = \sqrt{2}+1$$

Also $\sin 22.5° = \dfrac{\sqrt{2-\sqrt{2}}}{2},\ \cos 22.5° = \dfrac{\sqrt{2+\sqrt{2}}}{2}$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 4(7)

Question 42
How to prove identities in tangent and cotangent involving three angles?

Model Answer
If $A = B + C$: $\tan A = \dfrac{\tan B + \tan C}{1 - \tan B \tan C}$, cross-multiply: $\tan A - \tan B - \tan C = \tan A \tan B \tan C$.
Ex: $\tan 3A - \tan 2A - \tan A = \tan 3A \tan 2A \tan A$; $\tan 70° = \tan 20° + 2\tan 50°$ (since $\tan 70° \tan 20° = 1$).

Grade - Low
Practice map, main concept: Sheet: J-I 5; C-I 6 | DPP: 3(40)
Also used in: none

Question 43
How to write $|\sin A + \cos A|$ and $|\sin A - \cos A|$?

Model Answer

$$|\sin A + \cos A| = \sqrt{1+\sin 2A}, \qquad |\sin A - \cos A| = \sqrt{1-\sin 2A}$$

Remove the modulus with the sign from items 23, 24. Ex: $A \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$: $\sqrt{1-\sin 2A} = \sin A - \cos A$.

Grade - Medium
Practice map, main concept: Sheet: J-A 6; C-I 1 | DPP: 4(23-24)
Also used in: Sheet: C-I 4 | DPP: 4(2)

3A Formulae

Question 44

What is the formula for $\sin 3A$ in $\sin A$, and what is its application?

Model Answer

$$\sin 3A = 3\sin A - 4\sin^3 A \qquad \left(\sin^3 A = \frac{3\sin A - \sin 3A}{4}\right)$$

Use for cubes of sine and for equations in $\sin A$ of degree 3. Ex: $\sin^3 10°$ type sums; $\sin 3A = \frac{1}{2}$ gives a cubic in $\sin A$.

Grade - **High**

Practice map, main concept: Sheet: J-M 2; J-A 4; C-II 16, 29

Also used in: Sheet: J-I 25; C-I 3; C-II 21 | DPP: 3(32)

Question 45

What is the formula for $\cos 3A$ in $\cos A$, and what is its application?

Model Answer

$$\cos 3A = 4\cos^3 A - 3\cos A \qquad \left(\cos^3 A = \frac{3\cos A + \cos 3A}{4}\right)$$

Ex: $\cos\theta = \frac{1}{2}\left(a + \frac{1}{a}\right) \Rightarrow \cos 3\theta = \frac{1}{2}\left(a^3 + \frac{1}{a^3}\right)$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 9

Also used in: Sheet: C-I 13; C-II 16 | DPP: 3(33, 39)

Question 46

What is the formula for $\tan 3A$ in $\tan A$, and what is its application?

Model Answer

$$\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

Use for cubic equations in $\tan A$: $\tan A$, $\tan\left(A + \frac{\pi}{3}\right)$, $\tan\left(A + \frac{2\pi}{3}\right)$ are the roots of $t^3 - 3t^2\tan 3A - 3t + \tan 3A = 0$. Ex: $\tan A = \frac{1}{2} \Rightarrow \tan 3A = \frac{11}{2}$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: C-I 13 | DPP: 3(34)

6 Important Results

Question 47

What are the 6 important results?

Model Answer

$$\cot A - \tan A = 2\cot 2A, \qquad \cot A + \tan A = 2\operatorname{cosec} 2A$$

$$(1 + \tan A)(1 + \tan B) = 2 \quad \text{if } A + B = 45°$$

$$\sin\theta\sin(60° - \theta)\sin(60° + \theta) = \frac{1}{4}\sin 3\theta, \quad \cos\theta\cos(60° - \theta)\cos(60° + \theta) = \frac{1}{4}\cos 3\theta$$

$$\tan\theta\tan(60° - \theta)\tan(60° + \theta) = \tan 3\theta$$

Grade - **High**

Practice map, main concept: Sheet: J-II 2, 5-6; J-M 4, 10; C-II 15 | DPP: 3(7-8, 31-33, 35-36) | Question Bank: 7

Also used in: Sheet: J-II 3, 25, 40; J-M 1; C-II 34

Question 48
When do we apply the $\theta,\ 60^\circ-\theta,\ 60^\circ+\theta$ formulae?

Model Answer
When a product of three sines/cosines/tangents has angles $\theta,\ 60^\circ-\theta,\ 60^\circ+\theta$ (e.g. $20^\circ, 40^\circ, 80^\circ$).

Ex: $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \dfrac{\sqrt{3}}{8}$; $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 6 | DPP: 3(35-36)

Question 49
What to do when one term of the $\theta,\ 60^\circ-\theta,\ 60^\circ+\theta$ product is missing?

Model Answer
Multiply and divide by the missing factor (its value is often known, e.g. $\sin 60^\circ$), or note the missing angle is a standard one.

Ex: $\sin 10^\circ \sin 50^\circ \sin 70^\circ = \dfrac{\sin 30^\circ}{4} = \dfrac{1}{8}$; $\tan 10^\circ \tan 50^\circ \tan 70^\circ = \dfrac{1}{\sqrt{3}}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Value at 18° and 36°

Question 50
What is the value of $\sin 18^\circ$?

Model Answer

$$\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5}-1}{4}, \qquad \cos 18^\circ = \sin 72^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

From $\sin 2\theta = \cos 3\theta$ with $\theta = 18^\circ$: $4s^2+2s-1=0$. So $\operatorname{cosec} 18^\circ = \sqrt{5}+1$ is a root of $x^2-2x-4=0$.

Grade - High
Practice map, main concept: Sheet: J-II 34; J-M 9
Also used in: Sheet: J-II 6, 13; J-M 4; C-I 9, 12; C-II 9, 22

Question 51
What is the value of $\cos 36^\circ$?

Model Answer

$$\cos 36^\circ = \sin 54^\circ = \frac{\sqrt{5}+1}{4}, \qquad \sin 36^\circ = \cos 54^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$$

$\cos 36^\circ - \sin 18^\circ = \frac{1}{2}$, $\cos 36^\circ \sin 18^\circ = \frac{1}{4}$; $\sin 36^\circ$ is a root of $16x^4 - 20x^2 + 5 = 0$.

Grade - Medium
Practice map, main concept: Sheet: C-I 12
Also used in: Sheet: J-M 9

Componendo and Dividendo

Question 52

What is the result of componendo and dividendo?

Model Answer

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 53

When would we apply componendo and dividendo in trigonometry questions?

Model Answer

When a ratio of sines/cosines is given and a ratio of tangents/cotangents is asked (or vice versa); then use $C \pm D$.

Ex: $\sin 2A = \lambda \sin 2B \Rightarrow \dfrac{\tan(A+B)}{\tan(A-B)} = \dfrac{\lambda+1}{\lambda-1}$; $3\sin\beta = \sin(2\alpha+\beta) \Rightarrow \tan(\alpha+\beta) = 2\tan\alpha$.

Grade - Medium

Practice map, main concept: Sheet: J-I 19-21; J-II 12; J-M 6; C-I 3; C-II 7, 21

Also used in: none

General Formula for tan

Question 54

Write the general formula for tan of a sum and the pattern it follows.

Model Answer

$$\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

$$\tan(\theta_1 + \theta_2 + \cdots + \theta_n) = \frac{S_1 - S_3 + S_5 - \cdots}{1 - S_2 + S_4 - \cdots}$$

S_i = sum of products of the tangents taken i at a time: odd sums above, even sums below, signs alternate. If $A+B+C = \pi$ the numerator is 0.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 27

Continued Product

Question 55
What is the result of the continued product of a series of cosines and when do we apply it?

Model Answer

$$\cos\theta\cos 2\theta\cos 2^2\theta\cdots\cos 2^{n-1}\theta=\frac{\sin 2^n\theta}{2^n\sin\theta}$$

Apply when each angle is double the previous one (multiply and divide by $2\sin\theta$). Ex: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}=-\frac{1}{8}$; $\cos 36^\circ\cos 72^\circ=\frac{1}{4}$.
If angles are not doubling, reorder using $\cos(\pi-x)=-\cos x$ first.

Grade - Medium
Practice map, main concept: Sheet: J-II 15; J-M 3; C-II 10
Also used in: none

Trigonometric Identities for a Triangle

Question 56
What is the basic process of proving identities in sine and cosine for a triangle?

Model Answer
Combine two terms by $C\pm D$, then use $A+B=\pi-C$: $\sin(A+B)=\sin C$, $\cos(A+B)=-\cos C$, $\sin\frac{A+B}{2}=\cos\frac{C}{2}$; take the common factor and apply $C\pm D$ again.

Grade - Low
Practice map, main concept: Sheet: J-II 16-17; C-II 32 | DPP: 5(1, 3-7, 9-12)
Also used in: Sheet: C-II 2 | DPP: 5(2, 8, 13-14)

Question 57
What is the basic process of proving identities in tan and cot for a triangle?

Model Answer
Write $\tan(A+B)=\tan(\pi-C)=-\tan C$ (or $\cot\frac{A+B}{2}=\tan\frac{C}{2}$), expand and cross-multiply.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 58
Write the triangle identities for $\sum\sin 2A$, $\sum\cos 2A$, $\sum\sin^2 A$, $\sum\cos^2 A$.

Model Answer

$$\sum\sin 2A=4\sin A\sin B\sin C,\qquad\sum\cos 2A=-1-4\cos A\cos B\cos C$$
$$\sum\sin^2 A=2+2\cos A\cos B\cos C,\qquad\sum\cos^2 A=1-2\cos A\cos B\cos C$$

Also $\sum\sin A=4\prod\cos\frac{A}{2}$, $\sum\cos A=1+4\prod\sin\frac{A}{2}$.

Grade - Low
Practice map, main concept: DPP: 5(2, 8, 13-14)
Also used in: Sheet: C-I 2

Question 59
Write all 4 trigonometric identities for a triangle in tan and cot.

Model Answer

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C, \qquad \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

$$\sum \tan \tfrac{A}{2} \tan \tfrac{B}{2} = 1, \qquad \cot \tfrac{A}{2} + \cot \tfrac{B}{2} + \cot \tfrac{C}{2} = \cot \tfrac{A}{2} \cot \tfrac{B}{2} \cot \tfrac{C}{2}$$

They hold whenever $A + B + C = \pi$ (the first one also for $A + B + C = n\pi$).

Grade - Low
Practice map, main concept: Sheet: C-II 19, 31, 33
Also used in: Sheet: C-II 20

Question 60
What is the condition on triangle ABC if $\sum \sin^2 A = 2$?

Model Answer
$2 + 2\cos A \cos B \cos C = 2 \Rightarrow \cos A \cos B \cos C = 0$: the triangle is right-angled.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 61
What is the condition on triangle ABC if $\sum \cos^2 A = 1$?

Model Answer
$1 - 2\cos A \cos B \cos C = 1 \Rightarrow$ one angle is $90°$ - right-angled triangle.

Grade - Low
Practice map, main concept: Sheet: C-I 2
Also used in: none

Graphs

Question 62
Plot the graphs of all six trig functions and mark where they meet the x-axis.

Model Answer
$\sin x = 0$ at $x = n\pi$; $\cos x = 0$ at $x = (2n+1)\frac{\pi}{2}$; $\tan x$ zeros $n\pi$, asymptotes $(2n+1)\frac{\pi}{2}$; $\cot x$ zeros $(2n+1)\frac{\pi}{2}$, asymptotes $n\pi$.
$\sec x, \operatorname{cosec} x$ never meet the x-axis ($|y| \geq 1$). Periods: $\sin, \cos, \sec, \operatorname{cosec}$: 2π; $\tan, \cot$: π. Graphs are used to count solutions (points of intersection).

Grade - Medium
Practice map, main concept: Sheet: C-I 7 | Question Bank: 1
Also used in: Sheet: C-II 26

Range of Trigonometric Functions - Type 1

Question 63
What is the range of $\sin x, \cos x, \tan x, \cot x, \sec x$ and $\operatorname{cosec} x$?

Model Answer

$$\sin x, \cos x \in [-1, 1]; \quad \tan x, \cot x \in \mathbb{R}; \quad \sec x, \operatorname{cosec} x \in (-\infty, -1] \cup [1, \infty)$$

Grade - Low

Practice map, main concept: Sheet: C-I 18
Also used in: none

Question 64
How to find the range of the six functions when a condition on x is given?

Model Answer
Use the graph on the given interval (check monotonic pieces and end-points). Ex: $x \in \left[\frac{\pi}{6}, \frac{3\pi}{4}\right] \Rightarrow \sin x \in \left[\frac{1}{2}, 1\right]$, $\cos x \in \left[-\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}\right]$.

Grade - Low

Practice map, main concept: Sheet: J-I 27
Also used in: none

Question 65
What is the range of the squares of the six functions?

Model Answer

$$\sin^2 x, \cos^2 x \in [0, 1]; \quad \tan^2 x, \cot^2 x \in [0, \infty); \quad \sec^2 x, \operatorname{cosec}^2 x \in [1, \infty)$$

Grade - Medium

Practice map, main concept: Sheet: C-I 5 | Question Bank: 3
Also used in: none

Question 66
How to find the range of $af(x) + b$ when the range of $f(x)$ is known?

Model Answer
$f \in [m, M]$: if $a > 0$, $af + b \in [am + b, aM + b]$; if $a < 0$ the ends swap. Ex: $3 - 2\sin x \in [1, 5]$.

Grade - Low

Practice map, main concept: none
Also used in: Sheet: C-I 18

Question 67
How to find the range of $(f(x))^2$ when the range of $f(x)$ is known?

Model Answer
If $[m, M]$ contains 0: $f^2 \in [0, \max(m^2, M^2)]$; otherwise between m^2 and M^2. Ex: $\sin x - \frac{1}{2} \in \left[-\frac{3}{2}, \frac{1}{2}\right] \Rightarrow$ square $\in \left[0, \frac{9}{4}\right]$.

Grade - Medium

Practice map, main concept: none
Also used in: Question Bank: 3

Question 68

How to find the range of $\sqrt{f(x)}$ when the range of $f(x)$ is known?

Model Answer

Keep only the part $f \geq 0$: $f \in [m, M],\ M \geq 0 \Rightarrow \sqrt{f} \in [\sqrt{\max(m, 0)}, \sqrt{M}]$. Ex: $\sqrt{\sin x} \in [0, 1]$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 69

How to find the range of $\dfrac{1}{f(x)}$ when the range of $f(x)$ is known?

Model Answer

$f \in [m, M]$ of one sign: $\frac{1}{f} \in \left[\frac{1}{M}, \frac{1}{m}\right]$; if 0 is inside, split: $\left(-\infty, \frac{1}{m}\right] \cup \left[\frac{1}{M}, \infty\right)$. Ex: $\dfrac{1}{2 + \sin x} \in \left[\frac{1}{3}, 1\right]$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 23-24

Range of Trigonometric Functions - Type 2

Question 70

When to apply the concept of type 2?

Model Answer

When the expression becomes a quadratic in one function ($\sin x$, $\cos x$, $\tan x$, $\sin x + \cos x$...): complete the square and use the range of that function.

Ex: $\sin^2 x + \cos^4 x = 1 - \cos^2 x + \cos^4 x = \left(\cos^2 x - \frac{1}{2}\right)^2 + \frac{3}{4} \in \left[\frac{3}{4}, 1\right]$.

Grade - Low

Practice map, main concept: Sheet: J-I 24; J-II 19, 29

Also used in: Sheet: J-II 37

Question 71

How to solve type 2 questions with the help of quadratic equations?

Model Answer

Put $t = \sin x \in [-1, 1]$; $y = f(t)$ is a quadratic - find its min/max on $[-1, 1]$ (vertex if inside, else end-points). For $y = \dfrac{P(t)}{Q(t)}$ make it a quadratic in t and demand a root in the allowed interval.

Ex: $y = \sin^2 x - 3 \sin x + 2 = (t - \frac{3}{2})^2 - \frac{1}{4}$, $t \in [-1, 1] \Rightarrow y \in [0, 6]$.

Grade - Low

Practice map, main concept: Sheet: J-II 23, 35; C-I 15; C-II 8

Also used in: none

Range of Trigonometric Functions - Type 3

Question 72
What is the range of $y = a\sin x + b\cos x$? Derive it.

Model Answer

$$a\sin x + b\cos x = \sqrt{a^2+b^2}\,\sin(x+\phi),\ \tan\phi = \tfrac{b}{a}\ \Rightarrow\ y \in \left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$$

Ex: $3\sin x + 4\cos x + 2 \in [-3, 7]$.

Grade - Low
Practice map, main concept: Sheet: J-II 21-22, 30; C-II 18, 26
Also used in: Sheet: J-II 18, 24, 37; C-II 11, 14, 24

Question 73
When can we add the ranges of two functions directly?

Model Answer
Only when both reach their maxima (and minima) at the same x; otherwise the sum of ranges is only a bound. Ex: $\sin^2 x + \sin^4 x \in [0, 2]$ (both max at $x = \frac{\pi}{2}$), but $\sin x + \cos x \neq [-2, 2]$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 74
How to find the range of $y = a\sin^2 x + b\cos^2 x + c\sin x\cos x + d$?

Model Answer
Convert to $2x$: $y = \dfrac{a+b}{2} + d + \dfrac{b-a}{2}\cos 2x + \dfrac{c}{2}\sin 2x$, then type 3: $\dfrac{a+b}{2} + d \pm \dfrac{1}{2}\sqrt{(a-b)^2+c^2}$.
Ex: $3\sin^2 x + 5\cos^2 x + 2\sqrt{3}\sin x\cos x \in [2, 6]$.

Grade - Low
Practice map, main concept: Sheet: J-II 18, 24
Also used in: none

Question 75
How to find the range of $y = a\sin x + b\cos x$ when a condition on x is given?

Model Answer
Write $y = R\sin(x+\phi)$, find the interval of $x+\phi$ and read the range of sin on it.
Ex: $x \in \left[0, \frac{\pi}{2}\right]$: $\sin x + \sqrt{3}\cos x = 2\sin\left(x + \frac{\pi}{3}\right) \in [1, 2]$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 30

Question 76
What is the range of $y = |\sin x| + |\cos x|$?

Model Answer

$$y^2 = 1 + |\sin 2x| \in [1, 2]\ \Rightarrow\ y \in \left[1, \sqrt{2}\right]$$

Period $\frac{\pi}{2}$; maximum at odd multiples of $\frac{\pi}{4}$, minimum at multiples of $\frac{\pi}{2}$.

Grade - Low
Practice map, main concept: Sheet: J-II 31
Also used in: none

Range of Trigonometric Functions - Type 4

Question 77
What is the range of $y = a^2 \tan^2 x + b^2 \cot^2 x$?

Model Answer

$$y = (a \tan x - b \cot x)^2 + 2ab \;\Rightarrow\; y \in [2ab, \infty) \quad (a, b > 0)$$

Minimum when $\tan^2 x = \frac{b}{a}$ (always possible). Ex: $9 \tan^2 x + 16 \cot^2 x \in [24, \infty)$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 42

Question 78
What is the range of $y = a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x$?

Model Answer

$$y = (a^2 + b^2) + (a^2 \tan^2 x + b^2 \cot^2 x) \;\Rightarrow\; y \in \left[(a + b)^2, \infty\right)$$

Ex: $4 \sec^2 x + 9 \operatorname{cosec}^2 x \in [25, \infty)$.

Grade - Low
Practice map, main concept: Sheet: J-II 20
Also used in: none

Question 79
How to find the range of $y = a^2 \sin^2 x + b^2 \operatorname{cosec}^2 x$?

Model Answer
$y = (a \sin x - b \operatorname{cosec} x)^2 + 2ab$. Check whether $a \sin x = b \operatorname{cosec} x$, i.e. $\sin^2 x = \frac{b}{a} \leq 1$: if yes $y \in [2ab, \infty)$; if not, the least value is at $\sin^2 x = 1$: $y \in [a^2 + b^2, \infty)$.
Ex: $25 \sin^2 x + 9 \operatorname{cosec}^2 x \in [30, \infty)$; $9 \sin^2 x + 25 \operatorname{cosec}^2 x \in [34, \infty)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 80
How to find the range of $y = a^2 \cos^2 x + b^2 \sec^2 x$?

Model Answer
Same test: $\cos^2 x = \frac{b}{a}$ possible $(b \leq a) \Rightarrow y \in [2ab, \infty)$; otherwise $y \in [a^2 + b^2, \infty)$.
Ex: $\cos^2 x + 4 \sec^2 x \in [5, \infty)$.

Grade - Medium
Practice map, main concept: Sheet: J-M 11 | Question Bank: 10
Also used in: none

Summation of Series - Type 1

Question 81
What is the result of summation of series type 1 and how is it derived? What is the condition on the common difference?

Model Answer

$$\sum_{k=0}^{n-1}\sin(\alpha+k\beta)=\frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\sin\left(\alpha+\tfrac{(n-1)\beta}{2}\right),\quad \sum_{k=0}^{n-1}\cos(\alpha+k\beta)=\frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\cos\left(\alpha+\tfrac{(n-1)\beta}{2}\right)$$

Derivation: multiply by $2\sin\frac{\beta}{2}$, the terms telescope. Condition: $\beta\neq 2k\pi$. Memory: $\frac{\sin(n\cdot\frac{\text{c.d.}}{2})}{\sin\frac{\text{c.d.}}{2}}\times\sin\left(\frac{\text{first+last}}{2}\right)$.

Grade - Low

Practice map, main concept: Sheet: C-II 34

Also used in: Sheet: C-II 10, 12

Question 82
When would we apply summation of series type 1?

Model Answer

Sum of sines or cosines whose angles are in A.P. (also alternating signs: write $-\cos x=\cos(\pi-x)$ or $\cos(\pi+x)$).

Ex: $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=-\frac{1}{2}$; $\sum_{k=1}^{n}\cos\frac{2k\pi}{n}=0$ for $n\geq 2$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 83
What to do when terms of type 1 appear in square or cube form?

Model Answer

Reduce the power first: $\sin^2 x=\frac{1-\cos 2x}{2}$, $\cos^2 x=\frac{1+\cos 2x}{2}$, $\sin^3 x=\frac{3\sin x-\sin 3x}{4}$, $\cos^3 x=\frac{3\cos x+\cos 3x}{4}$; then sum each A.P. part.

Ex: $\sin^2\frac{\pi}{8}+\sin^2\frac{3\pi}{8}+\sin^2\frac{5\pi}{8}+\sin^2\frac{7\pi}{8}=2$.

Grade - Low

Practice map, main concept: Sheet: C-II 12 | DPP: 4(17)

Also used in: none

Summation of Series - Type 2

Question 84
On which theme are questions of type 2 based?

Model Answer

Telescoping: write $T_r=f(r+1)-f(r)$; then $\sum_{r=1}^{n}T_r=f(n+1)-f(1)$ (all middle terms cancel).

Grade - Medium

Practice map, main concept: Sheet: J-I 25; J-II 25-28, 40; J-A 2; C-I 19 | Question Bank: 5

Also used in: Sheet: C-I 17

Question 85

Which formulae are used to split the general term as a difference of two terms?

Model Answer

$$\frac{\sin(A-B)}{\cos A\cos B}=\tan A-\tan B,\qquad \frac{\sin(A-B)}{\sin A\sin B}=\cot B-\cot A$$
$$\operatorname{cosec}2x=\cot x-\cot 2x,\qquad \tan x=\cot x-2\cot 2x$$

Also $\tan(A-B)=\dfrac{\tan A-\tan B}{1+\tan A\tan B}$ gives $\tan A\tan B=\dfrac{\tan A-\tan B}{\tan(A-B)}-1$.

Grade - Medium

Practice map, main concept: DPP: 2(5)

Also used in: Sheet: J-II 25-28, 40; J-A 2 | DPP: 3(9) | Question Bank: 5

Question 86

How to decide which term should be in the numerator for splitting in type 2?

Model Answer

Put sin of the difference of the two angles of the denominator in the numerator (multiply and divide by it; it is a constant).

Ex: $\displaystyle\sum_{r=0}^{44}\frac{1}{\cos r^\circ\cos(r+1)^\circ}=\frac{1}{\sin 1^\circ}\sum(\tan(r+1)^\circ-\tan r^\circ)=\frac{1}{\sin 1^\circ}.$

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 26

Inequalities in a Triangle

Question 87

How to find the minimum/maximum value of a trig expression for a triangle (objective approach)?

Model Answer

For a symmetric expression the extreme is at $A=B=C=\frac{\pi}{3}$. Results: $\sum\cos A\le\frac{3}{2}$, $\sum\sin A\le\frac{3\sqrt{3}}{2}$, $\prod\cos A\le\frac{1}{8}$, $\sum\tan^2 A\ge 9$ (acute), $\sum\cot^2 A\ge 1$, $\sum\tan^2\frac{A}{2}\ge 1$.

Proof tools: $x^2+y^2+z^2\ge xy+yz+zx$ with $\sum\cot A\cot B=1$; or $C\pm D$ and a quadratic in $\cos\frac{A-B}{2}$; AM-GM.

Grade - Low

Practice map, main concept: Sheet: C-II 20

Also used in: none

Question 88

How to check whether the value obtained is a maximum or a minimum?

Model Answer

Compare with a nearly degenerate triangle (e.g. $A\to\pi,\ B,C\to 0$ or $A=\frac{\pi}{2},\ B=C=\frac{\pi}{4}$): if that value is larger, the equilateral value is a minimum; if smaller, a maximum.

Ex: $\sum\cos A$: $\frac{3}{2}$ at equilateral, $\to 1$ at $A\to\pi$, so $\frac{3}{2}$ is the maximum.

Grade - Low

Practice map, main concept: none

Also used in: none

Trigonometric Substitution

Question 89

What to substitute when $x^2 + y^2 = r^2$?

Model Answer

$$x = r\cos\theta,\ y = r\sin\theta$$

Ex: $x^2 + y^2 = 9 \Rightarrow 3x + 4y = 15\sin(\theta + \phi) \in [-15, 15]$.

Grade - Low

Practice map, main concept: Sheet: C-II 14, 24

Also used in: none

Question 90

What should be kept in mind when we use trigonometric substitution?

Model Answer

Take the parameter interval so that every value of the variable is covered exactly once (e.g. $x = \tan\theta,\ \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; $x = \sin\theta,\ \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

Common choices: $\sqrt{a^2 - x^2} \to x = a\sin\theta$; $1 + x^2 \to x = \tan\theta$; $\dfrac{2x}{1 - x^2} \to \tan 2\theta$; $x + y + z = xyz \to$ $x, y, z = \tan$ of angles summing to π.

Grade - Low

Practice map, main concept: Sheet: C-II 27

Also used in: Sheet: C-I 8